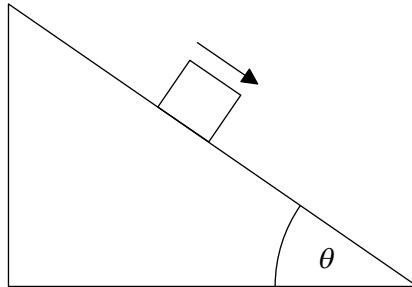


PAPER 1 / QUESTION 58

25 October 2023 / TZ2 (file label) / HL Paper 1 / original question 5 / 8823-6513

58. The diagram shows a block of mass m sliding at constant speed down an inclined plane. The plane makes an angle θ with the horizontal.



The coefficient of dynamic friction between the block and the plane is μ_D and the magnitude of the normal reaction force on the block is N . What is θ ?

- A. $\cos^{-1}\left(\frac{mg}{\mu_D N}\right)$
- B. $\cos^{-1}\left(\frac{\mu_D N}{mg}\right)$
- C. $\sin^{-1}\left(\frac{mg}{\mu_D N}\right)$
- D. $\sin^{-1}\left(\frac{\mu_D N}{mg}\right)$